TENECTEPLASE

High-alert drug

Single bolus

CHEMICAL NAME Recombinant tissue plasminogen activator variant (TNK-tPA) with 3 amino-acid substitutions (T103N, N117Q, KHRR296–299AAAA)

CLASS Fibrin-specific thrombolytic agent

TRADE NAMES Metalyse · TNKase

THERAPEUTIC USES / INDICATIONS

- Acute STEMI (within 6–12 h) when primary PCI not achievable in time; ideal for pre-hospital use.
- Acute ischaemic stroke (increasingly used, 0.25 mg/kg).

PHARMACODYNAMICS

- Binds fibrin and converts **plasminogen → plasmin** → clot lysis.
- More fibrin-specific → less systemic fibrinogen depletion.

ADMINISTRATION

- **IV single bolus over 5–10 seconds** — weight-based.
- Flush line before and after with saline (incompatible with dextrose).
- Give with aspirin, clopidogrel and anticoagulant (enoxaparin/heparin) in STEMI.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Minor bleeding (puncture sites).
- Nausea, hypotension.
- Fever.

ADVERSE EFFECTS (SERIOUS)

- Intracranial haemorrhage.
- Major bleeding.
- Reperfusion arrhythmias.
- Allergic reactions (rare).

DOSAGES

- **STEMI (weight-based, max 50 mg):** <60 kg 30 mg; 60–69 kg 35 mg; 70–79 kg 40 mg; 80–89 kg 45 mg; ≥90 kg 50 mg.
- **Age ≥75 yr:** consider half dose.
- **Ischaemic stroke:** 0.25 mg/kg (max 25 mg) bolus.

CONTRAINDICATIONS & CAUTIONS

- Same as alteplase: previous ICH, ischaemic stroke <3 months (STEMI), active bleeding, recent major surgery/trauma, suspected aortic dissection, severe uncontrolled hypertension, bleeding diathesis.

KEY INTERACTIONS

- **Anticoagulants, antiplatelets:** ↑ bleeding (used together by protocol in STEMI).
- **Dextrose solutions:** precipitates.

PHARMACOKINETICS

- **Onset:** immediate. **t½:** 20–24 min (initial) — allows single bolus.
- Cleared by liver.
- Greater fibrin specificity and PAI-1 resistance than alteplase.

NURSING MANAGEMENT

- Weigh/estimate weight accurately; select dose by band.
- Continuous ECG (reperfusion arrhythmias), BP, neuro status.
- Assess chest pain and ST resolution at 60–90 min (failed lysis → rescue PCI).
- Monitor puncture sites; minimise invasive procedures.
- Have defibrillator ready.

PATIENT EDUCATION

- Report headache, weakness, bleeding or chest pain return.
- Explain the need for transfer for angiography.

KEY POINTS

- Single-bolus thrombolytic.
- Flush — incompatible with dextrose.
- STEMI weight-banded dose, max 50 mg.
- Check ST resolution at 60–90 min.